

Course Code: CS301
Course Name: Operating Systems
Semester: Fall 2026

Course Objectives:

Understand the core concepts of Operating Systems, concurrency, and memory management.

Course Content:

UNIT 1: Processes

- Process Concept
- PCB
- Process States

UNIT 2: Scheduling

- FCFS
- SJF
- Round Robin

UNIT 3: Memory Management

- Paging
- Segmentation
- Virtual Memory

UNIT 4: Deadlocks

- Deadlock Characterization
- Banker's Algorithm
- Prevention

UNIT 5: File Systems

- File Concept
- Access Methods
- Directory Structure

Suggested Books:

- Primary Textbook: Silberschatz, Galvin, Gagne - Operating System Concepts
- Reference Book: Andrew S. Tanenbaum - Modern Operating Systems